latent2splat: Decoding 3D Gaussian Assets from Frozen Video-VAE Latents

Jason Lee Yubo Ruan Ryan Zhang

Stanford University CS231N

Overview

Can a frozen, general-purpose video-VAE latent be decoded directly into an explicit 3DGS asset?

- **In:** latent of a short orbit video (frozen LTX-2.3 VAE).
- **Out:** 3D Gaussians (mean, rot, scale, opacity, color).
- **Supervision:** multi-view render loss only.

Latent $(128, 2, 36, 24)$: $\sim$21$\times$ spatial compression, 9 frames $\to$ 2 slices. No full-res skips (unlike Splatter-Image).

Data

Split	Obj.	Cat.	Views
v7	1,365	171	55
v8	3,626	965	55
Comb.	4,991	1,136	55

Table 1. Objaverse-LVIS, 768×512, ≈275k frames (∼200 GB), Blender Cycles on Modal.

- Filter: manual ∼50% (v7); Gemini 2.5 Flash ∼37% (v8).
- 49 orbit + 6 canonical views; RGB, mask, Z-depth, cameras.
- Unit-sphere normalized; no augmentation.

Method

A — Strict decoder (`CleanGSDecoder`): 12-layer transformer (d768, 16h) → stride-2 upsample → $K{=}1$ ray-anchored Gaussian/pixel (98k–393k).

$$\mu = \mathbf{o} + \big(d_n + (d_f - d_n)\sigma(d)\big)\mathbf{r}$$

$$\mathcal{L} = w_{\text{fg}}\|\mathbf{M}\odot(\hat{\mathbf{I}} - \mathbf{I})\|_1 + 0.2(1-\text{SSIM})$$

$$+0.5\|\hat{\mathbf{A}} - \mathbf{M}\|_1 + 0.01\mathcal{L}_{\text{scale}} + \lambda_{\text{vgg}}\mathcal{L}_{\text{vgg}}$$

No global opacity penalty ⇒ avoids collapse.

B — Scaffolded fusion (teacher):

latent → LTX RGB → DA3 → fuse → 3DGS

9 views lifted with known cameras; support score, affine color cal, voxel fuse, alpha clean.

Final Model (∼1B)

`RGBDSurfaceTokenDecoder`: keep scaffold *evidence*, learn every *decision*. ≈1.5M surface tokens + 768-slot core (self-refine ×40 ≈ 850M); zero-init + gated ⇒ step 0 = scaffold exactly.

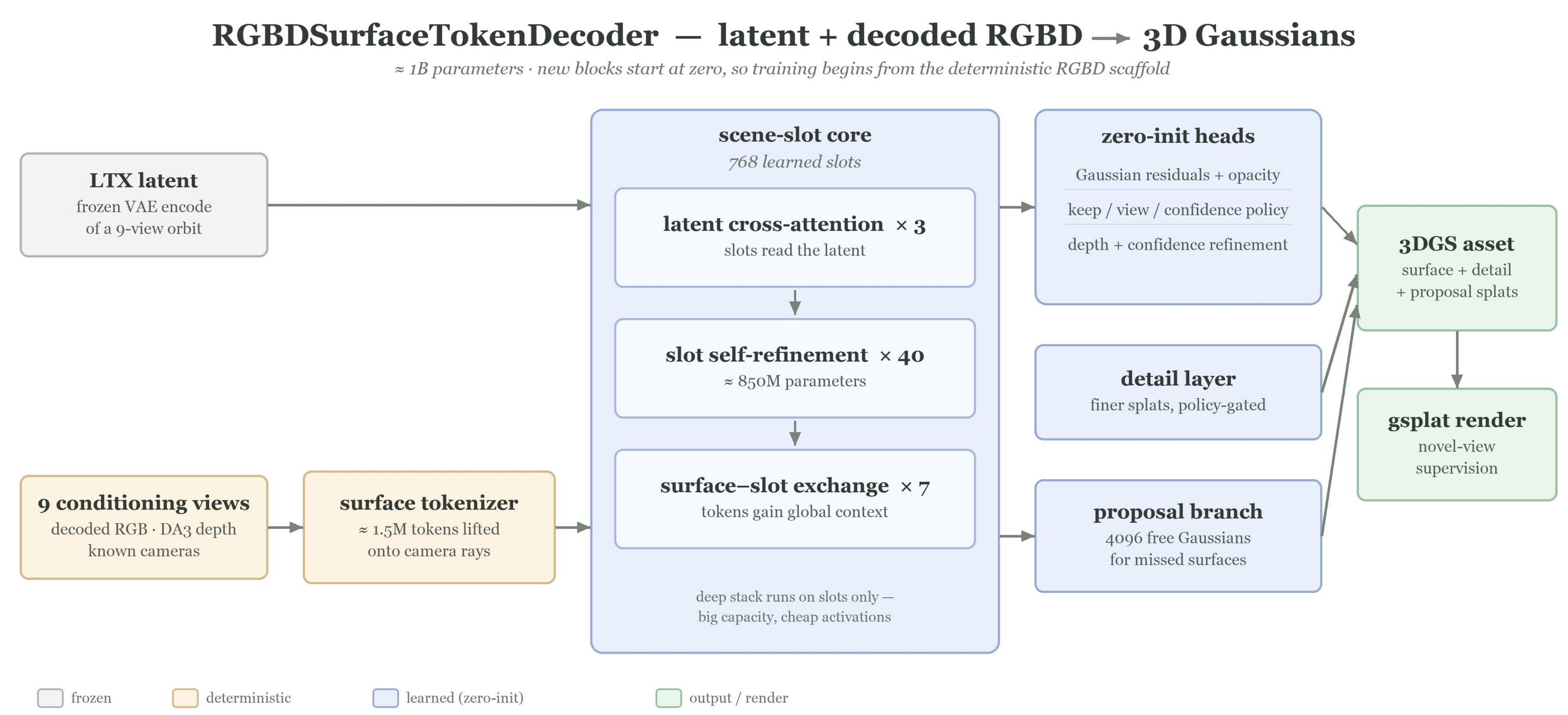

Figure 1. The ∼1B `RGBDSurfaceTokenDecoder`. RGBD surface tokens (≈1.5M) + LTX latent meet in a 768-slot core; zero-init gated heads refine, gate, and extend the splats. Step 0 reproduces the RGBD scaffold exactly.

Held-out Novel Views

Figure 2. Feed-forward 3DGS assets decoded from frozen LTX video-VAE latents — held-out objects, novel target cameras.

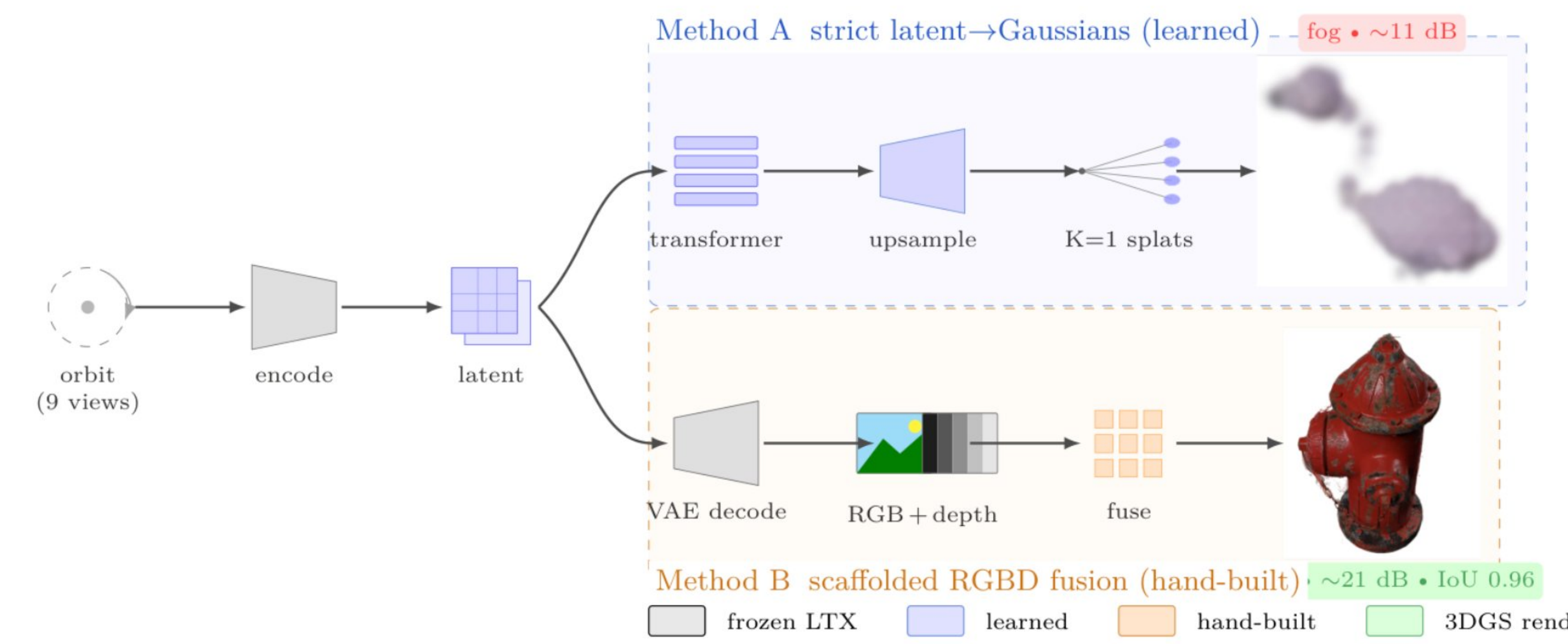

Figure 3. **A** strict (learned); **B** decode+fuse (hand-built). Gray frozen, blue learned, orange hand-built, green render.

Results

Strict (A)	Train	Held
Free fit (no net)	20.9	–
Decoder overfit	18.24	–
Multi-object (76)	14.48	11.30

Table 2. Reads 1 object; memorizes 76 → fog.

Scaffold (B)	FG	Shp	IoU
Target (48)	21.04	0.862	0.958
Surf. refine	21.14	0.835	0.959
Full (138)	20.65	0.946	0.958
Score gate (138)	20.71	0.942	0.957

Table 3. Fog gone: ∼21 dB, α-IoU ≈ 0.96.

Depth ablation: lower depth MAE ⇏ better render (-0.59 dB). Edge/multi-view consistency dominates.

Conclusion

- **Representation, not capacity.** $K{=}1$ ray anchoring can't densify/relocate.
- **Fog = equilibrium.** Translucent splats average to color; opacity gradient vanishes.
- **Latent not the wall.** Scaffold recovers ∼9 dB → content present; bottleneck is decoder + data.
- **Path forward:** learn the scaffold's evidence + visibility end-to-end.

yuboruan, rrzhang, lee2025 @stanford.edu